

Roots of $x^7 - 1$ in $\overline{\mathbb{F}}_2$

$$A = \mathbb{F}_2[x]/\langle x^7 - 1 \rangle$$

$$x^7 - 1 = (x+1)(x^3+x+1)(x^3+x^2+1)$$

Choose the generator polynomial

$$g(x) = x^3 + x + 1$$

Then

$$C = \langle \overline{g(x)} \rangle \subset A$$

is a cyclic code of length 7.

The roots split into Frobenius-orbits:

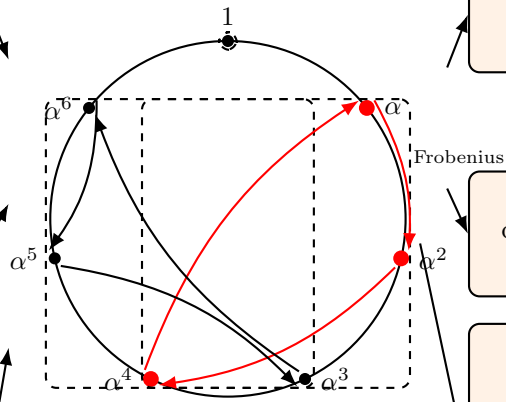
$$\{1\}, \quad \{\alpha, \alpha^2, \alpha^4\}, \quad \{\alpha^3, \alpha^5, \alpha^6\}.$$

Over $\mathbb{F}_2$, a polynomial cannot choose just one root from an orbit. It must choose the whole orbit.

So

$$g(x) = x^3 + x + 1$$

corresponds to the orbit

$$\{\alpha, \alpha^2, \alpha^4\}.$$


Frobenius acts by $z \mapsto z^2$,
so on exponents mod 7 it acts by $i \mapsto 2i$.

Cyclic-code root picture:
a generator polynomial $g(x) \mid (x^n - 1)$ selects Frobenius-orbits among the n -th roots of unity.

For this example,

$$g(x) = x^3 + x + 1 \iff \{\alpha, \alpha^2, \alpha^4\}.$$

ts of g .